integral substitution

integral substitution can only be used when there is anything other than an x in the input

ex:

there is something other than a x in the input

$$\int 2x \sin(x^2+1)\, dx$$

what ever is in the input that isn't an x will be your u that will be step 1 the second step will be take the derivative of u

$$\int 2x \sin(u)\, dx$$

$u = x^2+1$

$du = 2x\, dx$

$\frac{du}{2x} = dx$

after taking the derivative of u you will get dx by itself and plug in for dx

$$\int \cancel{2x} \sin(u) \frac{du}{\cancel{2x}}$$

after that take the integral and return u

$$\boxed{-\cos(x^2+1)+C}$$

$x^{1/2} \Rightarrow \frac{1}{2\sqrt{x}}$

$$\int \frac{\csc^2 \sqrt{x}}{\sqrt{x}}\,dx$$

$u = \sqrt{x}$

$du = \frac{1}{2\sqrt{x}}\,dx$

$2\sqrt{x}\,du = dx$

$$\int \frac{\csc^2 u}{\sqrt{x}}\,dx$$

$$\int \frac{\csc^2 u}{\cancel{\sqrt{x}}}\,2\cancel{\sqrt{x}}\,du$$

$$2\int \csc^2 u\,du \Rightarrow \boxed{-2\cot(\sqrt{x}) + C}$$

rember we are looking for an input the only input here is the $\csc^2\sqrt{x}$ so u would be $\sqrt{x}$ the denominator is Not an input

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$$\int \frac{\cot(\ln x)\cdot\csc(\ln x)}{x}\,dx$$

$u = \ln x$

$du = \frac{1}{x}\,dx$

$x\,du = dx$

$$\int \frac{\cot(u)\cdot\csc(u)}{\cancel{x}}\,\cancel{x}\,du$$

$$\int \cot(u)\cdot\csc(u)\,du \Rightarrow \boxed{-\csc(\ln x) + C}$$

$$\int \tan 3x \sec 3x \, dx$$

$u = 3x$

$du = 3\,dx$

$$\int \tan u \sec u \, \frac{du}{3}$$

$\frac{du}{3} = dx$

$$\frac{1}{3}\int \tan u \sec u \, du = \frac{1}{3}[\sec u]$$

$$\boxed{\frac{1}{3}\sec(3x) + C}$$

$$\int (x-1)(5x^2-10x)^{10}\,dx$$

$u = 5x^2 - 10x$

$du = 10x - 10\,dx$

$\frac{du}{10x-10} = dx$

$$\int \cancel{(x-1)}(u)^{10}\frac{du}{10\cancel{(x-1)}}$$

$$\frac{1}{10}\int (u)^{10} \Rightarrow \frac{(u)^{10+1}}{10+1} = \frac{1}{10}\left[\frac{(u)^{11}}{11}\right]$$

$$\boxed{\frac{(5x^2-10x)^{11}}{110} + C}$$

$(x-1)$ is not an input $(x-1)$ and $x-1$ are the same thing the input is $(5x^2-10x)$ because it is being raised to the power of 10 therefore making it an input